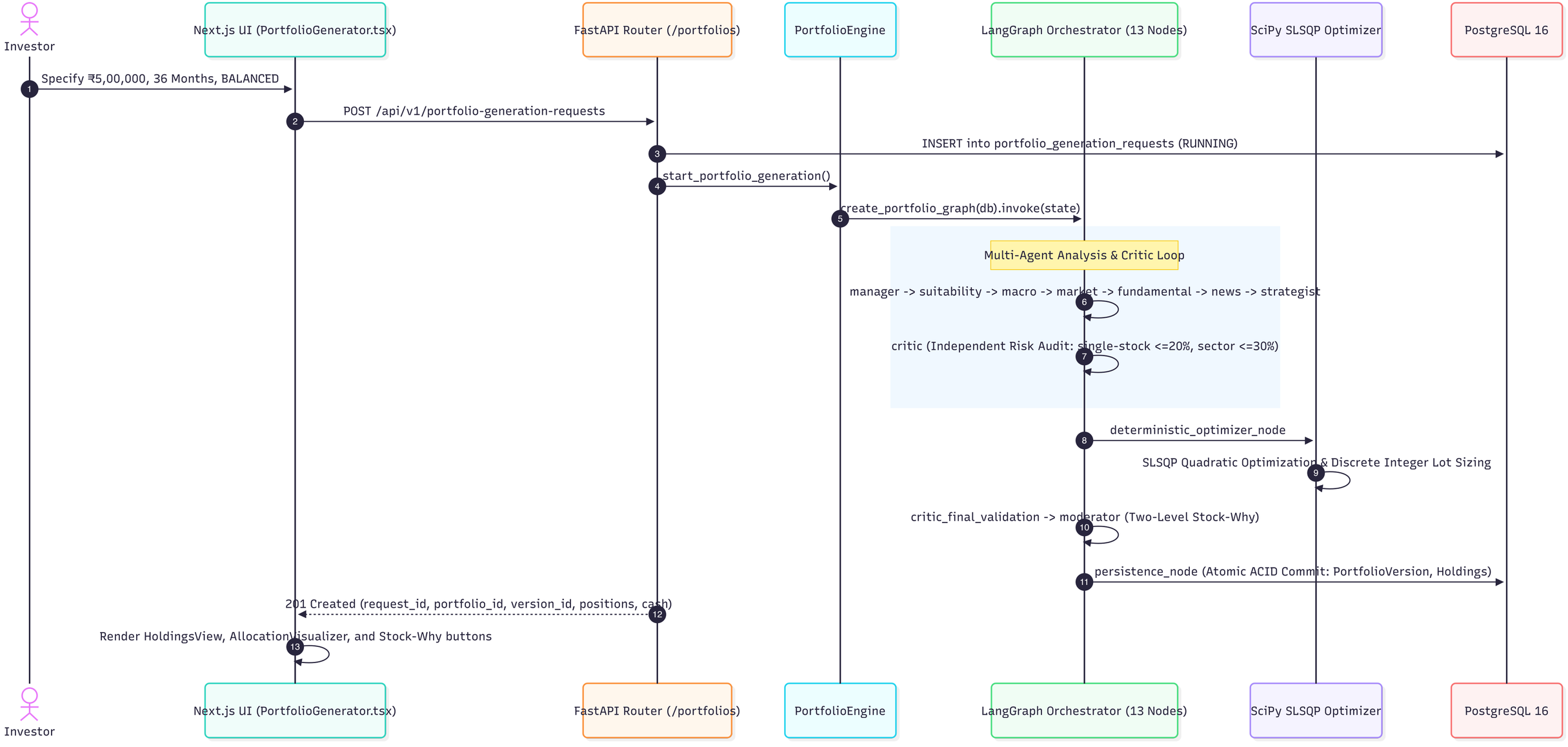

Investor
Next.js UI (PortfolioGenerator.tsx)
FastAPI Router (/portfolios)
PortfolioEngine
LangGraph Orchestrator (13 Nodes)
SciPy SLSQP Optimizer
PostgreSQL 16
1
Specify ₹5,00,000, 36 Months, BALANCED
2
POST /api/v1/portfolio-generation-requests
3
INSERT into portfolio_generation_requests (RUNNING)
4
start_portfolio_generation()
5
create_portfolio_graph(db).invoke(state)
Multi-Agent Analysis & Critic Loop
manager -> suitability -> macro -> market -> fundamental -> news -> strategist
6
critic (Independent Risk Audit: single-stock <=20%, sector <=30%)
7
8
deterministic_optimizer_node
SLSQP Quadratic Optimization & Discrete Integer Lot Sizing
9
critic_final_validation -> moderator (Two-Level Stock-Why)
10
11
persistence_node (Atomic ACID Commit: PortfolioVersion, Holdings)
201 Created (request_id, portfolio_id, version_id, positions, cash)
12
Render HoldingsView, AllocationVisualizer, and Stock-Why buttons
13
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